Project 1

General Instructions

- Before you begin coding, determine where you are starting and where you need to go. Map out a plan for your code. If you use AI to help with the code, write psuedo-code or specs. Submit your code plan and specs as a word document. Include a brief summary of how you used AI (if you use AI).
- Projects should be written in R and the report produced in RMarkdown (or Quarto if you already know & prefer it)
- Note what version of R and what package versions you used
- Provide well-documented commentary on your code throughout your project & generally follow good programming practices
- Ensure your final report looks profession, produces well designed tables, etc. Tables should not go onto a second page unless that is unavoidable (long table).
- Hide your code in your final output document
- Projects are due October 15, 2026 at 5pm EDT.

Background

Women with active ovarian cancer have chemotherapy appointments every two to three weeks. Physicians are concerned about patients visiting the emergency department (ED) between their chemo appointments and want to be able to predict at the end of the chemo appointment if the patient likely visit the ED or to be hospitalized (also called "Unanticipated hospital admission" or UHA) before the next appointment. The clinical team enlists the help of a biostatistician to create a predictive model. Before creating a predictive model, the data needs to be in a usable form. The goal of your project is to process the data to create a usable analytic dataset.

You can find the project data in the Project 1 assignment on Canvas. You will find two .csv files. One of these files contains *patient-level* data, i.e., data about the patients that are constant (each row is one patient). The other file is *encounter-level* data, i.e., data from office and hospital visits which varies over time (each row is one patient-encounter). An "encounter" is any contact a patient has with her medical care team. For this project, encounters are limited to office visits (chemo appointments), ED visits, and hospitalizations.

Patient data:

MRN	DOB	race	financialclass	ethnicity	hypertension	CHF	diabetes
DH1301	6/30/1957	Other	Private	non-Hispanic	N	N	N
JV9469	10/30/1964	White	Private	non-Hispanic	Y	N	N
TH8119	1/11/1981	White	Medicare	non-Hispanic	N	N	N
TJ3799	3/12/1949	White	Medicare	non-Hispanic	Y	N	Y
HP1319	9/3/1988	White	Private	non-Hispanic	Y	N	N

Encounter data:

MRN	contact_date	enc_type	temp	distress_score	WBC	BMI
HJ9754	6/26/2016	Office visit	97.91	2	4.87	28.33
GE5166	8/8/2016	Office visit	99.03	2	6.54	38.22
XV9573	1/20/2018	Office visit	99.15	2	7.64	32.13
CQ9338	7/5/2015	Office visit	99.09	3	6.94	25.09
DH1301	3/25/2018	Office visit	99.18	3	15.24	33.41

Project Components

Your project should include the following:

- Import both datasets. Do not hardcode a path!
- Merge the patient level data into the encounter level data
- This dataset will be your base "analytic dataset." Write a few sentences describing your dataset, including:
  - Granularity (what does each row represent?)
  - Dimensions (how many variables, how many rows?)
  - General description of variables (demographics, etc.)
- Address any data cleaning issues you see (missing data, implausible data, etc.)
- White blood cell count (WBC) is measured as counts of cells x 10^3 per $/mu$L. This patient population may include women with very low counts. Any count <1 in our data is under the limit of detection. Convert any value <1 to 1. (Aside, this does artificially lower our variance and you can learn about how to deal with that in another class.)
- Re-categorize WBC (white blood cell count) into a categorical variable with the following levels:
  - Low (<3.2)
  - Normal (3.2-9.8)
  - High (>9.8)
  - Not Taken
- Create a table of the categorical WBC variable (Make sure order the table rows logically, not alphabetically). This table should include the counts (%) of encounters within each WBC group.
- Create & print a table of the counts & percentages of race, ethnicity, financial class, hypertension, congestive heart failure, and diabetes at the patient level. A table like this is often called a Table 1.